

REWEAR Adversarial Stress-Test — Is This the Best Possible Concept for the Cox "Play With Purpose" / Carter's "Making & Remaking" Track?

TL;DR

- **REWEAR is the right thematic concept, but it is currently mis-fused. The highest-ceiling, most defensible, most prize-optimal version of this idea is NOT "designed urethanase + compliance triage classifier" — it is "co-designed cleavable elastane + de novo carbamate hydrolase," a single integrated molecular-design loop where Engine 1 designs the FIBER, Engine 2 designs the ENZYME that uniquely cleaves it, and the compliance classifier is demoted to a thin deployment wrapper (~10% of effort), not a co-equal engine.**
- The "first de novo carbamate hydrolase / urethanase" flag is confirmed OPEN as of June 2026. RFdiffusion2 (Nature Methods 2025, DOI 10.1038/s41592-025-02975-x) demonstrated serine, cysteine, and metallo-hydrolases; RFdiffusion3 (bioRxiv 10.1101/2025.09.18.676967, released open-source December 3, 2025) demonstrated cysteine hydrolases on 4-MU-butyrate; Bennett et al. (bioRxiv 10.1101/2025.11.21.689808, Nov 2025) extended to cysteine proteases on peptide bonds — but NO RFdiffusion-class paper has experimentally tested a carbamate substrate, and every urethanase advance to date (Li 2025 Adv Sci; ACS Catal 2025 phylogeny mining; Świderek 2025 JACS; Zheng 2026 ACS Catal "Computational Redesign") is structure-guided mutagenesis or metagenome mining, NOT de novo. The "AI-generative-designed enzymatically-cleavable elastane" flag is also open: every parallel industry effort (Fashion for Good "Stretching Circularity"; the Materiom/Ellen MacArthur Foundation cohort of 46 bio-based stretch fiber innovators, of which only ~4% are reported recyclable per materiom.org; Lycra Planet Agenda; Celanese × Under Armour NEOLAST; Hyosung's 2026 bio-BDO process) is bio-feedstock substitution or selective-dissolution chemistry — none are generative-AI-designed segmented-PU copolymers. (bioRxiv + 8)
- **Final ranking for this team / prize / sponsor:** (1) **REWEAR-FUSED** (the upgraded version specified below) — the only concept that simultaneously hits maximum technical ceiling, maximum "shouldn't be possible in 72h" shock, maximum sponsor fit, and maximum Cox Cleantech commercial-deployability fit; (2) AI-designed enzyme cocktail for whole-garment depolymerization — loses on Carter's fit (Carter's products are cotton + 2-5% elastane, not exotic blends) and on per-enzyme defensibility; (3) De novo PFAS-defluorinase for children's-textile finishes — extraordinarily high shock factor but loses on track theme ("Making & Remaking" = circularity, not detoxification); (4) Status-quo REWEAR as currently proposed — loses to REWEAR-FUSED on defensibility and "stapled-together" risk.

Key Findings

1. The compliance classifier (Engine 2 in the current REWEAR proposal) is a strategic liability, not an asset.

On the surface it looks deployable — it directly addresses the post-collection bottleneck in KIDCYCLE, which today only downcycles to insulation and pet bedding. But three problems make it a net negative for the win condition:

- **It is exactly the kind of artifact every competing team will build.** TrusTrace's CPSIA platform, Eurofins' Chem-ST methodology, and a dozen sustainability-compliance SaaS startups already do AI-enhanced chemical risk triage on supplier data. A 72-hour hackathon classifier built on CPSC Recalls API + EU Safety Gate + Peaslee PFAS data is a credible but not shocking artifact — it is structurally on the "AI computer-vision sorting / DPP triage" cluster the team's own avoid-list flagged. Cox's own 2026 Accelerator cohort already includes Kara, "a sustainability compliance platform powered by AI that transforms how investors and businesses manage sustainability risk" — Cox judges have just funded this exact archetype. (Trustrace + 2)
- **It dilutes the narrative.** Judges evaluating "shouldn't be possible in 72 hours" will discount a fused project where one engine is order-of-magnitude more audacious than the other. The classifier signals "we hedged."
- **It is a "two projects stapled together"** — exactly the risk the team itself flagged. The biological link between "designed enzyme that cleaves elastane" and "ML classifier that flags PFAS-contaminated garments" is purely narrative ("two walls"), not technical. A hostile judge will pull this thread.

The fix is not to delete the compliance layer entirely — Cox Cleantech explicitly funds deployable circular-economy tech (Cox-backed Nexus Circular runs a chemical recycling business for hard-to-recycle plastics; the 2026 Cleantech Accelerator cohort includes Silvis Materials for bio-based adhesives). The fix is to demote compliance to a ~10% software wrapper that handles intake-stream eligibility for the bio-engine, framed as "the bio-engine's safety gate," not as a co-equal engineering artifact. (Cox Enterprises) (Cox Enterprises)

2. The "single-urethanase-on-existing-elastane" framing leaves enormous ceiling on the table.

Even granting that the de novo urethanase is genuinely first, three structural critiques of Engine 1 as currently scoped:

- **Existing elastane is a hostile substrate.** Commercial Lycra/spandex is a segmented polyurethane-urea with N-aryl O-alkyl urea linkages, crystalline hard-segment domains

derived from MDI/TDI, and a soft polyether/polyester block. The Li 2025 Adv Sci paper, the ACS Catal 2025 phylogeny-guided mining paper, and the Świderek 2025 JACS mechanism paper all converge on the conclusion that natural and structure-guided-engineered urethanases have weak activity on these substrates. The Bornscheuer/Wei group's best two-step chemoenzymatic process requires ~200 °C glycolysis pretreatment before any enzyme touches the polymer. A 72-hour de novo enzyme on commercial Lycra is, in expectation, a low-activity in-silico artifact even if computationally beautiful. Judges who know the field will know this. (Wiley Online Library) (PubMed)

- **The polymer is fixed; only the enzyme is in the design loop.** This is the strategic mistake. Inverting the problem — *co-designing a new elastane variant with enzymatically pre-cleavable carbamate placement, then designing the matched enzyme* — turns a weakly-tractable depolymerization into a strongly-tractable one, AND opens a second open flag (AI-generative-designed cleavable elastane) that has no current claimant per the prior-art check.
- **There is a third engine hiding in plain sight: a substrate-enzyme co-evolution loop.** If the substrate is also a design variable, the team can run a joint inverse-design loop (polymer space × enzyme space) that no published paper has demonstrated for textile fibers. This is the "single bigger idea that subsumes both engines."

3. Cox Cleantech's commercial-deployability rubric strongly favors the upgraded concept.

Cox Cleantech's stated portfolio thesis (coxcleantech.com) is explicit: "renewing essential plastic resources, producing renewable energy and creating sustainable materials for the low carbon and circular economy." Their flagship cleantech investment Nexus Circular "converts used plastics — especially hard-to-recycle films and foams that would otherwise be bound for landfills and oceans — into clean, high-quality circular oil." The 2026 Accelerator cohort's Silvis Materials is precisely a "designed sustainable material" play (bio-based formaldehyde-free adhesive). The Residency (per the May 14, 2026 launch announcement) explicitly recruits "founders emerging from university research labs, incubators and accelerators in Georgia and neighboring states" — a deep-tech residency that rewards molecular-level innovation, not SaaS triage. REWEAR-FUSED maps perfectly onto this thesis as "a designed circular fiber chemistry plus the matched biocatalyst that closes its loop." The compliance-classifier version of REWEAR maps onto Kara, which is already in the 2026 cohort, and adds no differentiation. (Cox Cleantech + 3)

4. Carter's-specific fit dramatically favors the upgraded concept.

Carter's KIDCYCLE today shreds garments and downcycles. Per The Newnan Times-Herald (citing TerraCycle's official statement): "The first four-year partnership, from April 2021 to March 2025, collected 39.6 tons of clothing." Carter's added an in-store Zero Waste Box pilot on May 1, 2025; per the same TerraCycle statement: "Since launching the in-store KidCycle Zero Waste Box

program on May 1, Carter's has collected 4,407 pounds, or roughly 2.2 tons." Material outputs go to "home insulation and stuffing in workout equipment and furniture." The narrative gap Carter's needs filled is precisely "from used baby onesie to wearable baby onesie." Every alternative concept (engineered consortia, designed PETase variants, designed dye enzymes) fits this narrative loosely. REWEAR-FUSED fits it tightly: it directly proposes the molecular fiber Carter's could specify on a future onesie SKU AND the molecular catalyst that recycles it. That is a one-sentence partnership pitch for Carter's ESG team — and arguably a one-sentence pitch for the Cox Cleantech Residency interview. [\(The Newnan Times-Herald ...\)](#)

Details

PART 1 — Adversarial Alternatives Scan

I evaluated six frontier alternatives outside the avoid-list. Capsule assessments:

A. AI-designed PET-degrading enzyme variant (de novo, beyond FAST-PETase / LCC-ICCG). Ceiling: high. Shock: medium-low. Defensibility: WEAK — Carbios broke ground on its commercial 50 kt/y enzymatic PET recycling plant in Longlaville, France on April 25, 2024 (per Carbios's groundbreaking press release: "The plant will have a capacity of 50kt/year of prepared waste when operating at full capacity"); commissioning is now targeted for first-half 2028 after a 6–9 month construction delay disclosed in H1 2025 results, but the technical race is widely understood as over. Bhattacharya et al. 2025 (FEBS J., DOI 10.1111/febs.70228) just published an in-silico engineered LCC-ICCG variant; Ding et al. 2025 published an ML-driven novel polyester-hydrolase design. Buildability in 72h: high (mature toolchain). Carter's fit: medium (Carter's cotton products often have polyester thread, but the headline substrate is elastane, not PET). **Verdict: lower ceiling because the flag is mostly closed.** [\(Bioshine Enzymatic + 4\)](#)

B. AI-designed enzyme COCKTAIL for whole-garment depolymerization (urethanase + cellulase + cutinase + nylonase orchestrated by ML). Ceiling: very high. Shock: high. Defensibility: medium (Knott et al. 2020 PNAS established the multi-enzyme paradigm for PET; Frontiers in Systems Biology 2025 paper proposed enzyme-displaying *B. subtilis* spores for mixed PET/cotton; Nature Communications 2025 paper on *Rhodococcus jostii* + *Acinetobacter baylyi* consortium for mixed plastics). Buildability in 72h: low-medium — designing one de novo enzyme is hard, designing four is essentially impossible. The team would fall back on "we designed one enzyme + ML-optimized cocktail of existing ones," which is structurally a weaker claim. Carter's fit: medium-high. **Verdict: technically more ambitious than current REWEAR but lower buildability and lower defensibility per enzyme; strong runner-up.** [\(nih + 2\)](#)

C. Engineered microbial consortium or cell-free system for textile depolymerization. Ceiling: high. Shock: medium (consortia are an established direction). Defensibility: weak — the Sun et al. 2025 Nature Communications paper on engineered *Rhodococcus* + *Acinetobacter*, the 2025 Frontiers paper on spore-displayed enzymes, and the Canadian Journal of Microbiology 2025 review (DOI 10.1139/cjm-2025-0053) all show this is a crowded, well-funded field. Buildability in 72h: low (wet-lab biology is impossible in 72h; team would be limited to in-silico genome-scale metabolic models). **Verdict: defensibility too weak.** (PubMed Central)

D. AI-generative design of next-gen inherently-recyclable elastane (no enzyme). Ceiling: high. Shock: high. Defensibility: HIGH (per prior-art check — open flag). Buildability in 72h: medium-high (POLYT5, polyBART, GA+VFS, the Yu et al. Adv. Sci. 2025 vitrimer VAE (DOI 10.1002/advs.202411385) are all open-source; the team has to extend them to segmented copolymers, since polyBART's authors explicitly note "polyBART is currently limited to homopolymers and does not yet support the representation or generation of copolymers, polymer blends, or polymer composites/formulations"). Carter's fit: high. **Verdict: a strong alternative — but in isolation it loses the "designed enzyme" wow factor and the closed-loop story. This is the upgrade path INTO REWEAR-FUSED.** (arXiv)

E. ML catalyst/solvent discovery for blended children's-textile chemical recycling. Ceiling: medium-high. Shock: medium. Defensibility: medium (Azevedo et al. December 2025 DMSO-DBN selective elastane removal at moderate T, DOI 10.3390/polym17243247; Boschmeier TU Wien 2023–2025 work; Wang McGill 2022 LiCl/DMAc dissolution; binuclear zinc catalyst for polycotton in ACS Sus. Chem. Eng., DOI 10.1021/acssuschemeng.4c02547; IBM VolCat patent literature). Buildability in 72h: high (active learning over public reaction datasets). Carter's fit: medium. **Verdict: solid Tier-2 idea but lacks the biological shock factor that maximizes the prize.** (nih) (ACS Publications)

F. De novo enzyme that degrades a textile DYE/FINISH (azo dye reductase, PFAS C-F bond cleaver, formaldehyde-finish hydrolase). Ceiling: high for PFAS C-F (genuinely hard); medium for azo. Shock: very high for a PFAS-defluorinase. Defensibility: PFAS-defluorinase is a partially open flag (microbial defluorinases are an active area, but de novo design has not been demonstrated for textile-finish PFAS chemistries). Buildability in 72h: medium. Carter's fit: very high — California's Safer Clothes and Textiles Act (AB 1817, signed by Gov. Gavin Newsom on September 29, 2022) bans PFAS in textiles above 100 ppm Total Organic Fluorine from January 1, 2025, stepping down to 50 ppm from January 1, 2027; New York's ECL 37-0121 and Maine's LD 1503 add converging bans. **Verdict: this is the surprising dark-horse contender, but it sacrifices the "closed-loop" narrative — you don't recycle a garment by removing its PFAS — so it loses to REWEAR-FUSED on the track theme ("Making & Remaking").** However, "designed PFAS-

defluorinase" should be held in reserve as a pivot option (see Recommendations §6) or framed as a Phase-2 module within REWEAR-FUSED. SGS + 2

PART 2 — Head-to-Head

Axis	REWEAR (current)	REWEAR-FUSED (upgraded)	B. AI Enzyme Cocktail	D. AI- Designed Cleavable Elastane only	F. De Novo PFAS Defluorinase
Technical ceiling	High	Highest (joint polymer-enzyme co-design)	Very high	High	Very high (C-F cleavage)
72h shock	High	Highest (two designed molecules + closed loop)	High	Medium-high	Very high
Defensibility / open flag	High	Highest (two open flags fused into one)	Medium	High	Medium-high
72h buildability	Medium	Medium (harder but compositional)	Low-medium	Medium-high	Medium
Carter's fit	High	Highest (literally proposes a new Carter's fiber)	Medium	Medium-high	High (PFAS in kidswear)
Cox Residency fit	Medium-high	Highest (new material + biocatalyst = Nexus Circular-shaped story)	Medium	Medium-high	Medium
"Stapled together" risk	High (two engines, weak link)	Low (one design loop)	Low	Low	Low

Winner on every axis: REWEAR-FUSED. The only axis where current REWEAR ties is technical ceiling, and even there the joint co-design loop strictly dominates a single enzyme.

PART 3 — REWEAR Upgrade Paths

The single transformation that maximizes the prize: **fuse the two engines into one molecular co-design loop. Recast the project as the "first computationally co-designed circular fiber +**

biocatalyst pair," not as "enzyme + classifier."

Concretely, the 72-hour build becomes:

1. **Engine A — Cleavable elastane generative design.** Use polyBART/POLYT5 or a custom GNN-VAE/GA hybrid (extended to segmented block copolymers — this extension is itself a publishable methodological contribution) to generate candidate segmented PU-urea variants with: (i) carbamate placement engineered to be enzymatically accessible (no crystalline hard-segment occlusion); (ii) Tg and elastic modulus targets matched to spandex-equivalent stretch ($T_g < -50\text{ }^{\circ}\text{C}$ soft block, $T_g > 150\text{ }^{\circ}\text{C}$ hard block, elongation at break $\geq 400\%$); (iii) feedstock constraint to bio-based monomers (Hyosung's announced 2026 bio-BDO, isosorbide, succinic acid). Output: a top-10 ranked library of candidate cleavable-elastane molecules with predicted mechanical properties and ranked enzymatic-accessibility scores. (Sport-Tek)
2. **Engine B — Matched de novo carbamate hydrolase.** For each top candidate elastane, extract the carbamate microenvironment, build the theozyme (anchored on the Paiva 2025 Chem Sci / Świderek 2025 JACS transition state but generalized over the candidate substrate's specific R-group geometry), run RFdiffusion3 (released open-source December 3, 2025 per the IPD announcement) for atom-level scaffolding, LigandMPNN for sequence design, Boltz-2 for structure + affinity confirmation, and PLACER for transition-state geometry validation. Output: ≥ 3 de novo enzyme candidates per top elastane with predicted kcat/KM, computational success rates against the AME (atomic motif enzyme) benchmark of 41 catalytic sites, and downloadable PDB files. (Nature) (bioRxiv)
3. **Engine C — Compliance wrapper (10% effort).** A thin web service that screens incoming KIDCYCLE garments against the public datasets the team already identified (CPSC Recalls, EU Safety Gate, Peaslee/Toxic-Free Future PFAS), and triages them as "REWEAR-eligible / lab-test / divert." This becomes the front-door story for the Carter's pilot — "the AI router that decides which onesies flow into the molecular recycling line" — not the technical centerpiece.
4. **Demo artifact.** A live "submit your garment label" web app that, on demand: (a) classifies the garment; (b) shows the proposed cleavable-elastane replacement molecule with predicted properties; (c) shows the matched designed enzyme with structural visualization (Mol* / NGL Viewer); (d) animates the closed loop ("today: shredded $\rightarrow$ insulation; with REWEAR: depolymerized $\rightarrow$ re-spun $\rightarrow$ new onesie").

Why the upgrade dominates

- It collapses the "two stapled projects" risk to zero: there is one inverse-design problem with two coupled variables.
- It uses ALL the tools the team already chose (RFdiffusion3, LigandMPNN, Boltz-2, PLACER) AND adds a second open-source toolchain (polyBART/POLYT5/vitrimer-VAE)

without changing the team's expertise profile.

- It claims TWO open novelty flags simultaneously ("first de novo carbamate hydrolase" + "first AI-generative co-designed cleavable elastane fiber"), each defensible, with mutually reinforcing narratives.
- It is the only concept that lets the team say, on stage, "Carter's, here is the molecule for your next onesie, and here is the enzyme that closes its loop — neither existed 72 hours ago."

Race-timing and scoop risk

The Baker lab pattern of releases through 2025 (serine hydrolase Lauko 2025; cysteine hydrolase RFD2/RFD3; cysteine protease Nov 2025 Bennett et al. preprint at bioRxiv 10.1101/2025.11.21.689808) strongly suggests carbamate hydrolases are the next obvious de novo target. The C&EN coverage of the December 2025 RFdiffusion releases ("A trio of AI methods tackles enzyme design," cen.acs.org) notes that designed-enzyme success rates remain low and that experimental screening is required. The team has a window of roughly 6–18 months on the urethanase flag and roughly 12–24 months on the cleavable-elastane flag before plausible scooping (Fashion for Good "Stretching Circularity," Materiom/EMF cohort, or a Baker-lab/Bornscheuer collaboration could reach either flag). For a June 17, 2026 demo, both flags are open. (C&EN + 2)

Recommendations

Primary recommendation (decision-grade): Switch from REWEAR-as-currently-proposed to REWEAR-FUSED as defined above.

1. **By kickoff (June 14, 2026):** lock in the four-person role split as (i) Polymer-AI lead (extends polyBART/POLYT5 or the Yu et al. vitrimer-VAE to segmented block copolymers; trains on the segmented-PU and Materiom literature); (ii) Enzyme-AI lead (runs the RFdiffusion3 + LigandMPNN + Boltz-2 + PLACER pipeline conditional on substrates emitted by the Polymer-AI lead); (iii) Full-stack / demo lead (builds the live submission web app, the visualization layer with Mol* or NGL Viewer, and the compliance triage wrapper); (iv) Storyteller / Carter's-pitch lead (owns the "from your old onesie to your next onesie" narrative, the Carter's KIDCYCLE integration mockup, and the Cox Cleantech Residency commercialization plan).
2. **Hour 0–12:** ship the compliance wrapper end-to-end first (lowest-risk artifact; guarantees a working demo even if the molecular pipelines partially fail). This is a hackathon-grade hedge, not a strategic centerpiece.
3. **Hour 12–48:** run the two co-design loops in parallel. Generate $\geq 10,000$ candidate elastane molecules; filter to ≤ 100 by predicted mechanical/cleavability scores; emit substrate microenvironments to the enzyme pipeline; generate $\geq 1,000$ backbones per candidate substrate; filter to top-10 via Boltz-2 affinity + PLACER transition-state geometry.

4. **Hour 48–66:** integrate. Build the live demo. Pre-compute the top 3 substrate-enzyme pairs for the demo so nothing is dependent on real-time GPU during the pitch.
5. **Hour 66–72:** rehearse the pitch. Lock the "two open flags, one closed loop" narrative. Prepare for hostile judge questions: prior-art distinctions (vs. UMG-SP2 family, vs. Lauko 2025 serine hydrolase, vs. Carbios PETase, vs. Celanese × Under Armour NEOLAST recyclable elastoester, vs. Fashion for Good "Stretching Circularity"), 72h scope honesty (which steps are in-silico predictions vs. validated), and the 24-month commercialization roadmap explicitly tied to the Cox Residency \$10K stipend + 6-week mentorship.

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6. **Benchmarks that would change the recommendation:**
 - If, between now and June 14, a paper is published demonstrating a de novo RFdiffusion-class carbamate hydrolase (watch bioRxiv for "RFdiffusion" + "carbamate"/"urethane" weekly; watch ipd.uw.edu and bakerlab.org news feeds): pivot Engine B to a designed PFAS-defluorinase or a designed urea hydrolase for polyamide-elastane blends.
 - If a paper is published demonstrating AI-generative co-design of a cleavable segmented polyurethane fiber: pivot Engine A to a designed dye-receptor enzyme pair (matched de novo azo-reductase + designed cleavable azo-anchored chromophore), preserving the "co-designed pair" novelty.
 - If Fashion for Good "Stretching Circularity" announces a commercial-scale AI-designed elastane between now and June 14: pivot the framing from "first in the world" to "first open-source, sponsor-deployable" and lean harder on the Carter's KIDCYCLE integration narrative.

Caveats

- **The single biggest risk to REWEAR-FUSED is not technical but rhetorical: judges may not believe the team's in-silico outputs without wet-lab validation.** The mitigation is to (i) report computational success rates against published benchmarks (AME for enzymes, polymer property prediction MAE for fibers); (ii) cite the Lauko et al. 2025 / Nature Methods evidence that RFdiffusion2-designed enzymes have catalytic proficiencies comparable to natural enzymes after experimental testing of fewer than 96 sequences per design (per the Nature Methods abstract: "We further design enzymes for three distinct catalytic mechanisms and identify active candidates after experimentally testing fewer than 96 sequences in each case"); and (iii) explicitly offer the Cox Residency pitch as "we want the \$10K stipend and 6 weeks to validate this in a Georgia Tech / Emory wet lab" — which is exactly what the Residency is designed for. Nature
- **Polymer-AI for segmented copolymers is genuinely an open methodological problem; polyBART's own authors note their model "is currently limited to homopolymers and does not yet support the representation or generation of copolymers, polymer blends, or**

polymer composites/formulations." The team must be honest that their copolymer extension is a hackathon-grade prototype, not a peer-reviewed contribution. This is acceptable; the contribution is the joint loop, not the polymer-design model itself. (arXiv)

- **Boltz-2 affinity predictions have known limitations on novel chemistries.** Wan, Zhang, Xue & Coveney (UCL, submitted to arXiv March 2, 2026, "On the Reliability of AI Methods in Drug Discovery: Evaluation of Boltz-2 for Structure and Binding Affinity Prediction") conclude: "while Boltz-2 offers substantial speed for initial screening, it lacks the energetic resolution required for lead identification." The Synteny London paper (arXiv 2512.06592, Dec 2025) shows Boltz-2-PPI underperforms vs. sequence-based methods on protein-protein affinity. Use PLACER as the dominant geometry-of-catalysis validator and Boltz-2 as a secondary affinity filter; report uncertainty intervals, not point estimates. (arXiv) (arXiv)
- **The team should not over-promise on Carter's deployability.** A new elastane chemistry takes 5-10 years to commercialize at fiber-mill scale; the honest framing is "open-source design files + matched enzyme as a research artifact Carter's can license or co-develop," not "ready for your next product drop."
- **Carbios's commercial PET plant is not yet operational.** Per Carbios's H1 2025 results, construction was delayed 6-9 months after the April 25, 2024 groundbreaking; commissioning is now targeted for first half 2028. The team should not point to Carbios as "operating" — it is "constructing." This actually strengthens the REWEAR-FUSED narrative: enzymatic textile recycling at scale is still genuinely an open commercial opportunity, not a settled one. (Intelligent Living)
- **The current proposal's framing of the compliance classifier as a co-equal engine should be discarded entirely. Keep the compliance functionality; demote the framing.**